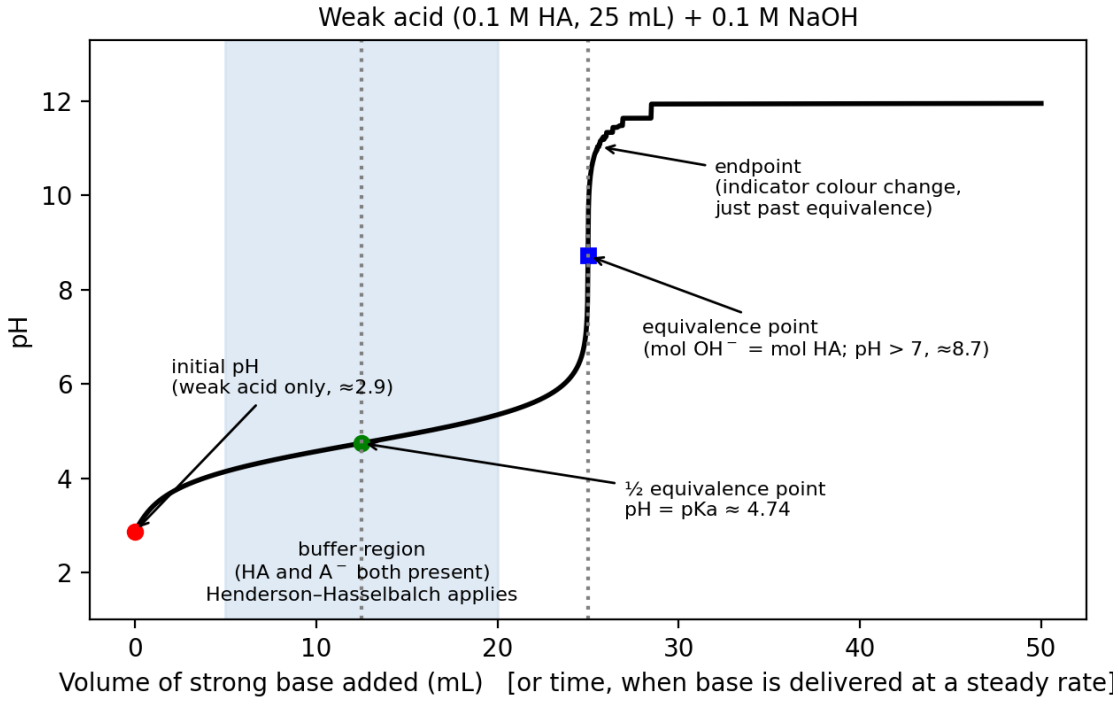
CHEM 151 Lab — Completed Pre-Lab Exercises

Pre-lab work from all 11 manuals, answered. Sections 1 to 4 are written pre-lab exercises (Acid-Base, Kinetics I, Le Chatelier, Intermolecular Forces). Section 5 is Making Solutions and Dilutions, whose required pre-lab calculations I missed on the first pass. Sections 6 to 8 are not labeled pre-lab in the manuals but contain tables and calculations you can prepare before lab (Equilibrium Constant, Electrochemistry predictions, Freezing Point Depression predictions). Kinetics II and Thermochemistry have no pre-lab and nothing computable in advance. Copy answers into your notebook by hand, and check any that depend on your instructor's conventions.

1. Acid-Base Chemistry

Q1. Sketch the weak acid / strong base titration curve and label it.



Key points (example: 25 mL of 0.1 M HA, $K_a = 1.8 \times 10^{-5}$, titrated with 0.1 M NaOH):

- *Initial pH*: only the weak acid is present, so $[H^+] = \sqrt{K_a C}$ and pH ≈ 2.9.
- *Buffer region*: from the first drops of base until about 90 % of the equivalence volume. HA and A^- coexist.
- *½ equivalence point*: half the HA has been converted to A^-, so $[HA] = [A^-]$ and pH $= pK_a$. This is how you read K_a off the curve.

- *Equivalence point*: moles of base added = moles of acid originally present. Only A^- remains, which is a weak base, so pH > 7 (about 8.7 here).
- *Endpoint*: the point where the indicator or probe signals a change. It is just past equivalence and is not the same thing as the equivalence point, though it is close for a well-chosen indicator such as phenolphthalein.

Q2. Henderson–Hasselbalch equation.

$$\mathrm{pH} = \mathrm{p}K_a + \log \frac{[\mathrm{A}^-]}{[\mathrm{HA}]}$$

It applies in the **buffer region**, the rising section before the equivalence point and roughly between 10 % and 90 % neutralized. It is exact at the half-equivalence point, where the log term is zero. It does not apply at the initial point (no A^- yet), at the equivalence point (no HA left), or after it (excess OH^- sets the pH).

Q3. Mass needed for 100 mL of 0.1 M. Moles needed = 0.1 M × 0.100 L = 0.0100 mol, so mass = 0.0100 mol × molar mass.

Compound	Molar mass (g/mol)	Mass for 100 mL of 0.1 M
NaCl	22.99 + 35.45 = 58.44	**0.584 g**
NH_4Cl	14.01 + 4(1.008) + 35.45 = 53.49	**0.535 g**
NaF	22.99 + 19.00 = 41.99	**0.420 g**

Q4. Net ionic equations and prediction. Each salt dissociates completely. Then check whether either ion reacts with water.

NaCl: $NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$. Na^+ is the cation of a strong base and Cl^- is the anion of a strong acid. Neither reacts with water. **Neutral** (pH ≈ 7).

NH_4Cl: $NH_4Cl(s) \rightarrow NH_4^+(aq) + Cl^-(aq)$, then $NH_4^+(aq) + H_2O(l) \rightleftharpoons NH_3(aq) + H_3O^+(aq)$. NH_4^+ is the conjugate acid of a weak base and Cl^- does nothing. **Acidic** (pH < 7).

NaF: $NaF(s) \rightarrow Na^+(aq) + F^-(aq)$, then $F^-(aq) + H_2O(l) \rightleftharpoons HF(aq) + OH^-(aq)$. F^- is the conjugate base of the weak acid HF. **Basic** (pH > 7).

2. Kinetics of H_2O_2 Decomposition, Part I

Table 1 is given in the manual (all mixtures total 60.0 mL: 15.0 + 35.0 + 10.0, 15.0 + 25.0 + 20.0, 30.0 + 20.0 + 10.0). Use $M_1V_1 = M_2V_2$, i.e. $M_2 = M_1V_1/V_{\text{total}}$.

Sample: Mixture 1 KI, (0.10)(15.0)/60.0 = 0.025 M. Mixture 1 H_2O_2, (0.88)(10.0)/60.0 = 0.147 M.

Table 2. Final concentrations

Component	Mixture 1	Mixture 2	Mixture 3
$[H_2O_2]$	**0.147 M**	**0.293 M**	**0.147 M**
$[KI] = [I^-]$	**0.025 M**	**0.025 M**	**0.050 M**

What the design shows: Mixture 2 doubles only H_2O_2 (compare with 1), and Mixture 3 doubles only I^- (compare with 1). If rate doubles, the order is 1. If it quadruples, the order is 2. If it does not change, the order is 0.

3. Le Chatelier's Principle

Balanced net ionic equations for each compound dissolved in water:

Compound	Equation
K_2CrO_4	$K_2CrO_4(s) \rightleftharpoons 2\,K^+(aq) + CrO_4^{2-}(aq)$
H_2SO_4	$H_2SO_4(aq) \rightarrow H^+(aq) + HSO_4^-(aq)$ (strong, complete), then $HSO_4^-(aq) \rightleftharpoons H^+(aq) + SO_4^{2-}(aq)$
NaOH	$NaOH(s) \rightarrow Na^+(aq) + OH^-(aq)$
CH_3COOH	$CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq)$ (weak, partial)
$NaCH_3COO$	$NaCH_3COO(s) \rightarrow Na^+(aq) + CH_3COO^-(aq)$
HCl	$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$ (strong, complete)
NH_4Cl	$NH_4Cl(s) \rightleftharpoons NH_4^+(aq) + Cl^-(aq)$

Strong electrolytes use a single arrow in most textbooks, while the manual's example uses a double arrow for the dissolving equilibrium. Follow whichever your instructor wants. You can write H_3O^+ in place of H^+.

4. Intermolecular Forces

Table 1. Pre-Lab Table

Substance	Formula	Structural formula	Molar mass (g/mol)	H-bond?
ethanol	C_2H_5OH	CH_3–CH_2–O–H	46.07	Yes
1-propanol	C_3H_7OH	CH_3–CH_2–CH_2–O–H	60.10	Yes
1-butanol	C_4H_9OH	CH_3–CH_2–CH_2–CH_2–O–H	74.12	Yes
n-pentane	C_5H_{12}	CH_3–CH_2–CH_2–CH_2–CH_3	72.15	No
methanol	CH_3OH	CH_3–O–H	32.04	Yes
n-hexane	C_6H_{14}	CH_3–$(CH_2)_4$–CH_3	86.18	No

Draw each structure out fully in the notebook, with every C–H bond shown and the O–H bond drawn explicitly on the alcohols. The H-bond column is "Yes" only where H is bonded directly to O (or N

or F). Alkanes have only C–H bonds, so they have dispersion forces only. Dispersion forces grow with molar mass, which explains the trend among the alcohols and between pentane and hexane.

5. Making Solutions and Dilutions

Molar masses: $Fe(NO_3)_3 \cdot 9H_2O$ = 55.845 + 3(62.004) + 9(18.015) = 404.00 g/mol. KSCN = 39.098 + 32.06 + 12.011 + 14.007 = 97.18 g/mol. Mass = M × V(L) × molar mass.

Part A: mass needed for each stock solution (100 mL, solvent is the nitric acid solution)

Solution	Moles needed	Mass
a. 0.1 M Fe^{3+} from $Fe(NO_3)_3 \cdot 9H_2O$	0.1 M × 0.100 L = 0.0100 mol	0.0100 × 404.00 = **4.040 g**
b. 20 mM Fe^{3+}	0.020 M × 0.100 L = 0.00200 mol	0.00200 × 404.00 = **0.808 g**
c. 20 mM SCN^- from KSCN	0.00200 mol	0.00200 × 97.18 = **0.194 g**

The older version of the manual adds *d.* 250 mL of 0.12 M $S_2O_3^{2-}$ from $Na_2S_2O_3 \cdot 5H_2O$ (248.18 g/mol), in distilled water: 0.12 × 0.250 = 0.0300 mol, so **7.445 g**.

Part B: dilutions, using $M_1V_1 = M_2V_2$ so $V_1 = M_2V_2/M_1$

Solution	Calculation	Stock to use	Nitric acid solution to add
a. 50 mL of 8 mM Fe^{3+}	(8)(50)/20	**20.0 mL**	30.0 mL
b. 50 mL of 6 mM Fe^{3+}	(6)(50)/20	**15.0 mL**	35.0 mL
c. 50 mL of 4 mM Fe^{3+}	(4)(50)/20	**10.0 mL**	40.0 mL
d. 100 mL of 0.2 mM SCN^-	(0.2)(100)/20	**1.00 mL**	99.0 mL

Fill the 50 mL (or 100 mL) volumetric flask to the mark with the nitric acid solution. The volume added is the final volume minus the stock volume.

Part C: tables for next week's Equilibrium Constant lab

Table 2 (calibration, 30.0 mL final each). SCN^- is diluted from 0.2 mM: $[SCN^-]_{in}$ = (0.2 mM)(V_{SCN})/30.0. Because SCN^- is limiting, $[FeSCN^{2+}]_{eq} = [SCN^-]_{in}$. The Fe^{3+} is the same in all four: (0.100 M)(10.0)/30.0 = 0.0333 M.

Solution	SCN^- volume	$[SCN^-]_{in}$ (M)	$[FeSCN^{2+}]_{eq}$ (M)
Blank	0.0 mL	0	0
1	5.0 mL	3.33×10^{-5}	3.33×10^{-5}
2	10.0 mL	6.67×10^{-5}	6.67×10^{-5}
3	20.0 mL	1.33×10^{-4}	1.33×10^{-4}

Table 3 (reaction solutions, 20.0 mL final each, 10.0 mL of each reagent, so every concentration is halved):

Solution	Fe^{3+} used	$[Fe^{3+}]_{in}$ (M)	$[SCN^-]_{in}$ (M)
4	10.0 mL of 4.00 mM	2.00×10^{-3}	1.00×10^{-4}
5	10.0 mL of 6.00 mM	3.00×10^{-3}	1.00×10^{-4}
6	10.0 mL of 8.00 mM	4.00×10^{-3}	1.00×10^{-4}

The Equilibrium Constant manual's Table 4 uses the same solutions (10.0 mL of 4.00, 6.00 and 8.00 mM Fe^{3+} plus 10.0 mL of 0.2 mM SCN^-).

6. Measuring an Equilibrium Constant (what can be prepared)

No written pre-lab, but the manual says to use dilution calculations to fill in Table 4, and the ICE setup can be done ahead. The initial concentrations are in Section 5, Part C.

ICE setup for solutions 4 to 6, with $x = [FeSCN^{2+}]_{eq}$ from the calibration line ($x = (A - b)/m$):

	Fe^{3+}	SCN^-	$FeSCN^{2+}$
I	$[Fe^{3+}]_{in}$	1.00×10^{-4} M	0
C	$-x$	$-x$	$+x$
E	$[Fe^{3+}]_{in} - x$	$1.00 \times 10^{-4} - x$	x

$$K_c = \frac{[FeSCN^{2+}]}{[Fe^{3+}][SCN^-]} = \frac{x}{([Fe^{3+}]_{in} - x)([SCN^-]_{in} - x)}$$

Upper bound for x: it cannot exceed 1.00×10^{-4} M, since SCN^- is limiting. Expect K_c in the low hundreds (literature is about 150 to 300 depending on conditions).

7. Electrochemistry (predictions for Tables 1 to 4)

No pre-lab is printed. These are the table entries worked out from the manual's Table 16.1, with $E^\circ_{cell} = E^\circ_{red}(\text{cathode}) - E^\circ_{red}(\text{anode})$. Confirm against what you observe.

Part A. Metals in 6 M HCl. Overall: $M(s) + 2H^+(aq) \rightarrow M^{2+}(aq) + H_2(g)$, with oxidation $M \rightarrow M^{2+} + 2e^-$ and reduction $2H^+ + 2e^- \rightarrow H_2$ ($E^\circ = 0.00$ V).

Metal	Oxidation half-reaction	E°_{ox} (V)	E°_{cell} (V)	Expected observation
Cu	$Cu \rightarrow Cu^{2+} + 2e^-$	-0.34	-0.34	No reaction (non-spontaneous)
Sn	$Sn \rightarrow Sn^{2+} + 2e^-$	$+0.1375$	$+0.1375$	Very slow, few bubbles
Fe	$Fe \rightarrow Fe^{2+} + 2e^-$	$+0.447$	$+0.447$	Steady slow bubbling
Zn	$Zn \rightarrow Zn^{2+} + 2e^-$	$+0.7618$	$+0.7618$	Faster bubbling

Metal	Oxidation half-reaction	E°_{ox} (V)	E°_{cell} (V)	Expected observation
Mg	$Mg \rightarrow Mg^{2+} + 2e^-$	+2.372	+2.372	Vigorous, fast

Ordering of reactivity (Mg > Zn > Fe > Sn > Cu) follows the reduction table: the more negative E°_{red}, the stronger the reducing agent. Cu sits above H^+ and does not react.

Part B. Aluminum can in $CuCl_2$. The scratched interior removes the Al_2O_3 layer.

- Oxidation: $Al \rightarrow Al^{3+} + 3e^-$, $E^\circ_{ox} = +1.662$ V (times 2)
- Reduction: $Cu^{2+} + 2e^- \rightarrow Cu$, $E^\circ_{red} = +0.34$ V (times 3)
- Cell: $2\,Al(s) + 3\,Cu^{2+}(aq) \rightarrow 2\,Al^{3+}(aq) + 3\,Cu(s)$, $E^\circ_{cell} = 0.34 + 1.662 =$ **+2.00 V**

Observation: copper-colored deposit on the scratched ring, and the can tears along it because the aluminum wall is eaten through.

Part C. Sodium in water.

- Oxidation: $Na \rightarrow Na^+ + e^-$, $E^\circ_{ox} = +2.71$ V (times 2)
- Reduction: $2\,H_2O + 2e^- \rightarrow H_2 + 2\,OH^-$
- Cell: $2\,Na(s) + 2\,H_2O(l) \rightarrow 2\,Na^+(aq) + 2\,OH^-(aq) + H_2(g)$

Using the manual's table (writing the reduction as $2H^+ + 2e^- \rightarrow H_2$, 0.00 V) gives $E^\circ_{cell} = +2.71$ V. In water, where the reduction is the one above ($E^\circ = -0.828$ V), it is +1.88 V. Either way it is strongly spontaneous. Phenolphthalein turns pink because OH^- is produced (basic), and the H_2 can ignite.

Part D. Zn/Cu cell.

- Anode (oxidation): $Zn(s) \rightarrow Zn^{2+} + 2e^-$, $E^\circ_{ox} = +0.7618$ V
- Cathode (reduction): $Cu^{2+} + 2e^- \rightarrow Cu(s)$, $E^\circ_{red} = +0.34$ V
- Cell: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$, $E^\circ_{cell} \approx$ **+1.10 V** (manual: 1.100 V; its Table 16.1 values give 1.10 V)
- Shorthand: $Zn(s)\,|\,Zn^{2+}(aq, 1\,M)\,\|\,Cu^{2+}(aq, 1\,M)\,|\,Cu(s)$, where $\|$ is the salt bridge (KNO_3).

Why the leads go as stated: the red (positive) lead goes on the metal with the larger reduction potential (Cu, the cathode) and black on Zn (the anode). The probe reports the potential of red relative to black, so this arrangement reads a positive voltage, consistent with the spontaneous direction. Reversed leads would read about −1.10 V.

NaOH additions, using $E = E^\circ - \dfrac{RT}{nF}\ln Q$ with $Q = \dfrac{[Zn^{2+}]}{[Cu^{2+}]}$:

- Adding OH^- to the $\mathbf{Cu^{2+}}$ side precipitates $Cu(OH)_2$, lowering $[Cu^{2+}]$. Q rises, so the voltage **drops** and the reaction becomes less spontaneous.
- Adding OH^- to the $\mathbf{Zn^{2+}}$ side precipitates $Zn(OH)_2$, lowering $[Zn^{2+}]$. Q falls, so the voltage **rises** and the reaction becomes more spontaneous. (With a lot of excess OH^-, the zinc hydroxide can re-dissolve as $Zn(OH)_4^{2-}$, so the effect has a limit.)

8. Freezing Point Depression (predictions)

No pre-lab is printed. These are the expected values to compare against, using $\Delta T_{fp} = K_{fp}\,m\,i$ with $K_{fp} = 1.86$ °C kg/mol for water.

Solute	Moles	Molality (mol/kg)	i (ideal)	Predicted ΔT_{fp}	Predicted T_{fp}
Sucrose, 17 g (342.30 g/mol) in 50 g water	0.0497	0.993	1	1.85 °C	−1.85 °C
NaCl, 2.5 g (58.44 g/mol) in 50 g water	0.0428	0.856	2	3.18 °C	−3.18 °C

The manual does not state the water mass for NaCl ("repeat Part D using 2.5 g"), so I assumed 50 g, the same as sucrose. If your water mass differs, redo the molality. At about 0.9 m, NaCl will measure below the ideal $i = 2$ (roughly 1.8), since the manual notes ions stop behaving independently at higher concentration. After the lab, $i = \Delta T_{fp}/(K_{fp}\, m)$.

9. Not included

Kinetics II (Table 1 is just Mixture 1 from Kinetics I, with the same concentrations: 0.147 M H_2O_2 and 0.025 M I^-) and Thermochemistry (all work is measured in lab) have nothing to prepare beyond reading. For Kinetics II, convert your three temperatures to kelvin before plotting $\ln k$ against $1/T$.